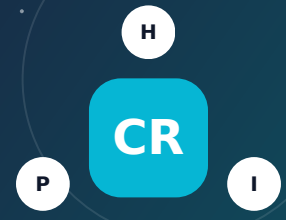


WATER AND EROSION HAZARD

Rivers, Streams & Creeks

A creek that looks small or dry can become a fast flood channel, erode a bank, or back up through a hidden culvert.



1 Understand the factor

What it is - and why it matters

Rivers, streams, and creeks include perennial channels, seasonal waterways, ephemeral washes, drainage ditches, and buried or culverted channels. Nearby water can provide habitat and recreation, but it also creates flood, erosion, bank-stability, groundwater, mosquito, and access considerations. Risk depends on watershed size, slope, rainfall intensity, upstream development, debris, channel capacity, levees, culverts, and the vertical elevation of the home.

How risk can reach a household

- Flash flooding can rise rapidly after intense rain upstream, sometimes before local rainfall appears severe.
- Bank erosion and channel migration can remove soil beneath yards, walls, roads, bridges, and foundations.
- Blocked culverts, bridges, and storm drains can redirect water into streets and properties outside mapped channels.
- High groundwater and seepage near channels can affect crawlspaces, basements, septic systems, and slope stability.

What people often miss

- A dry channel may carry the highest flows only during rare storms, when observation by a buyer is least likely.
- Some waterways run beneath streets or lots; a culvert failure can create sudden sinkholes or surface flow.
- Riparian buffers, maintenance easements, habitat rules, and permitting limits can restrict fences, trees, grading, and repairs.
- Flood risk may be driven by an upstream bottleneck or burn scar rather than the water visible beside the property.

? Five checks before buying

1

Review FEMA maps, local creek studies, watershed plans, culvert records, and historical aerial photographs.

2

Walk the site after rain if possible and inspect bank erosion, debris lines, undercut trees, sediment, seepage, and outfalls.

3

Determine who owns and maintains the channel, bank, culvert, levee, bridge, and access easement.

4

Check whether the home, driveway, utilities, septic system, and evacuation route are above expected flood levels.

5

Obtain flood and sewer-backup insurance quotes and ask about erosion, groundwater, and loss-of-use exclusions.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 06

H Health impacts

- Fast water can cause drowning, impact injuries, falls, electrocution, and vehicle entrapment.
- Persistent dampness and post-flood mold can worsen asthma and respiratory symptoms.
- Floodwater can contain sewage, animal waste, fuel, pesticides, pathogens, and sharp debris.
- Mosquitoes, odors, repeated warnings, and loss of access can affect sleep, outdoor activity, and mental health.

P Property impacts

- Flooding and erosion can damage foundations, retaining walls, landscaping, pools, fences, utilities, septic systems, and access roads.
- Sediment and debris can block drainage and cause repeated shallow flooding even without a major river overflow.
- Bank stabilization may require engineering, environmental permits, neighbor coordination, and expensive construction access.
- Waterfront appeal may be offset by maintenance duties, restricted use, insurance costs, and future channel change.

I Insurance implications

- Homeowners insurance generally excludes external flooding and often excludes erosion, groundwater, and gradual seepage.
- Flood insurance may cover direct building and contents damage but not all land, landscaping, decks, pools, or temporary housing.
- Sewer or drain backup often requires a separate endorsement and may not apply when the dominant cause is flood.
- Insurers may consider prior claims, elevation, repetitive-loss history, access, and local wildfire or flood conditions.

Possible long-term health implications of buying nearby

- Long-term health effects are mainly linked to repeated flooding, dampness, contaminated sediment, mosquitoes, and chronic worry.
- Children may have more contact with soil and water, so testing and hygiene are important after contamination events.
- Recurring nighttime flood alerts can disrupt sleep, while damaged crossings can delay emergency or medical access.
- A well-maintained, elevated home near a stable channel can have a very different risk profile from a low home beside an eroding bank.

Evidence lens

Flood behavior is well established, but local channel geometry, culverts, watershed changes, and extreme rainfall control parcel-level outcomes. Historical high-water evidence is especially valuable.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 06

V Verify the actual risk

- Identify mapped and hidden channels, flood zones, drainage easements, culverts, levees, and upstream bottlenecks.
- Ask public works and long-term neighbors about overtopping, road closures, erosion, and maintenance history.

P Protect the property

- Maintain legal drainage paths, vegetation, bank protection, gutters, sump systems, and backflow devices.
- Do not dump yard waste or build in a channel or easement; obtain permits before bank or culvert work.

M Monitor and respond

- Use rainfall and flood alerts, watch upstream conditions, and leave early when the only access route can flood.
- After flooding, avoid contact, photograph levels, remove wet materials, and test affected wells or septic systems.

I Prepare insurance records

- Carry flood coverage based on realistic building and contents values, even when purchase is not mandatory.
- Review sewer-backup, service-line, temporary-housing, erosion, and groundwater provisions in writing.

REAL-LIFE EXAMPLE

Waverly, Tennessee Flash Flood

August 21, 2021

Extreme rainfall caused catastrophic flash flooding in and around Waverly, Tennessee. The National Weather Service reports 20 deaths in Humphreys County. Water rose rapidly, swept through homes and roads, and left little time for evacuation in parts of the community.

Why it matters: The disaster illustrates how a relatively small creek and watershed can become life-threatening quickly and why upstream rainfall, warning time, and safe routes matter.

[Open official incident record - weather.gov >](#)**Questions to resolve before making a decision**

1. Review FEMA maps, local creek studies, watershed plans, culvert records, and historical aerial photographs.
2. Walk the site after rain if possible and inspect bank erosion, debris lines, undercut trees, sediment, seepage, and outfalls.
3. Determine who owns and maintains the channel, bank, culvert, levee, bridge, and access easement.
4. Check whether the home, driveway, utilities, septic system, and evacuation route are above expected flood levels.

Official sources and mapping tools[National Weather Service flash-flood statistics - weather.gov >](#)[USGS Water Data for the Nation - waterdata.usgs.gov >](#)[FEMA flood maps - fema.gov >](#)